



Sultan Qaboos University
College of Engineering
 Office of Assistant Dean for
 Industrial Training and Community Services

Weekly Report

Week #: 9

Date **28/06/2026**

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	Department: ☐ CAE ☐ ECE ☐ MIE ☐ PCE ☒ MCE	Mobile: 72471416
	Full Name: Saju Varuheoc	Email:
	Training Company & Department: Ministry of Defence – Engineering support	Tel./Mobile: 335314

Tasks Completed (Briefly Describe any tasks that you completed during this week):

Inspected 4-stroke diesel generator: combustion, cooling, ventilation, lubrication & electrical systems	Studied wet cylinder liner function (piston guidance, heat transfer, combustion sealing)
Assisted in cylinder liner chemical cleaning and O-ring groove scrubbing	Checked bare metal surfaces for cavitation pitting after cleaning
Observed Cummins V12 generator parameters: RPM, coolant temp, oil pressure, voltage & load	Reviewed AVR, circuit breaker panel wiring, and commissioning readiness

Tasks in Progress (if any):

Cylinder liner reassembly and new O-ring installation	Engine block torque procedure study
Fuel injection system testing preparation	

Plan for next week (if different from this week):	
• learn manufacturing engineering process	• observe workshop safety prosedure
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Problems/Challenges/Recommendations (if any):	
No major problem we have phased this week.	
Supervisor Comments (your feedback is very important for our continuing improvement):	
Satisfactory (Good Progress). Has a solid understanding of basic Diesel theory. Works safely and maintains a clean workspace which reflects positively in the shop environment. Grasps the function of major components (Pistons, Injectors, turbochargers) with more practice.	

<u>Student:</u> I have read and understood the whole content of the Students' Training Manual. Signature:	<u>Supervisor:</u> I will do my best to guide the student towards achieving all required deliverables. Signature:
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Note: A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. **Originals should be placed in the Appendix of the Final Report.**

Weekly Report

TRAINEE INFORMATION

Trainee Name	Malek Al Shukily
Student ID / National ID	141776
University / Institution	Sultan Qaboos University
Field of Study	Mechanical / Electrical Engineering
Training Department	Engineering Support — Generator Maintenance
Training Supervisor	Saju varuheoc
Report Period	Week 1 28 June 2026

WEEK 1 TRAINING OBJECTIVES

- Familiarize with the engineering support workshop environment and safety protocols
- Identify and describe major systems of the 4-stroke diesel generator set
- Observe and participate in cylinder liner cleaning and inspection procedures
- Study the operating parameters of a 12-cylinder heavy-duty diesel generator (Cummins)
- Understand the function of the alternator, AVR, and electrical control panel

ACTIVITIES PERFORMED

1. Diesel Generator Systems Inspection (4-Stroke)

A comprehensive visual inspection of a single-cylinder 4-stroke diesel generator was conducted. Key findings per system:

- Combustion System: Engine block, cylinder head, rocker arm cover, and Delphi fuel injection pump verified in good condition. No fuel leaks detected.
- Cooling System: Cooling water inlet/outlet connections and 8-blade plastic cooling fan confirmed serviceable; circuit prepared for pressure testing.
- Ventilation System: TOP-brand air filter connected via blue hose to intake manifold — clean and unobstructed.
- Lubrication System: Oil filter, camshaft gear, idle gear, and fuel pump gear all properly meshed and lubricated.
- Electrical / Alternator: Unit mounted on flywheel housing; AVR and circuit breaker panel wired and ready for commissioning.

2. Cylinder Liner Cleaning & Inspection

Wet cylinder liner maintenance was observed and assisted. Liners were submerged in a chemical wash bath; technicians manually scrubbed outer walls and O-ring grooves to remove rust, scale, and old sealant deposits. Post-cleaning, bare metal surfaces were inspected for cavitation pitting. New O-rings were staged for reinstallation.

3. 12-Cylinder Cummins Generator — Operational Study

An operational study of a Cummins V12 industrial diesel generator was undertaken, covering:

- 4-stroke diesel cycle: compression ignition, crankshaft rotation, and electromagnetic induction in the alternator stator/rotor.
- Engine parameters: oil pressure (3–5 bar), coolant temperature (80–95°C), engine speed (1500 RPM / 50 Hz).
- Electrical parameters: voltage, frequency, current (A), and load (kW/kVA) monitoring via the digital control panel.
- Cooling fan system: belt tension inspection, blade integrity check, and radiator core cleanliness verification.

SKILLS & KNOWLEDGE GAINED

- Practical identification of diesel generator sub-systems and their interdependencies
- Understanding of wet cylinder liner function: piston guidance, heat transfer, combustion sealing, and fluid separation
- Chemical cleaning technique for engine components and post-clean quality inspection
- Reading and interpreting generator operating parameters from a digital control panel
- Workshop safety awareness and tool handling under supervision

OBSERVATIONS & COMMENTS

The first week provided a strong foundational understanding of diesel power generation as applied in defence-critical infrastructure. The hands-on cylinder liner cleaning exercise illustrated how corrosion directly impacts cooling efficiency and reinforced the importance of scheduled preventive maintenance. The study of the Cummins V12 highlighted the precision required to maintain stable electrical output for sensitive military equipment. The engineering support team demonstrated exemplary professionalism and were forthcoming in knowledge transfer.